

Name:

Teacher: Assessment No.



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

Mathematics Extension 1

Year 11 Yearly Examination, 2023

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen.
- Calculators approved by NESA may be used.
- The HSC Reference Sheet will be provided.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.

	Marks
Section I: Multiple Choice	/10
Section II: Question 11	/15
Section II: Question 12	/15
Section II: Question 13	/15
Section II: Question 14	/15
Total	/70

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

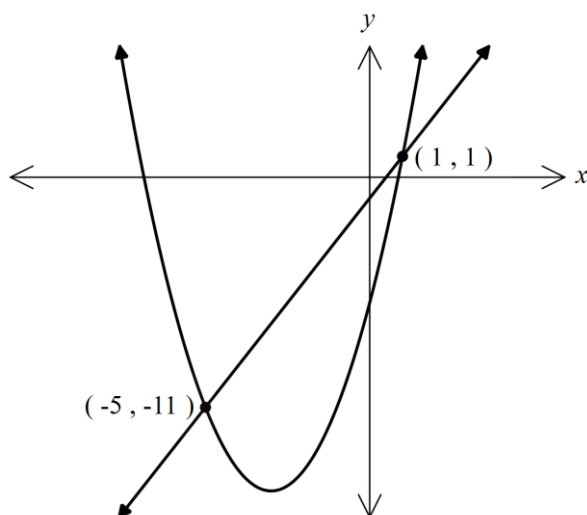
Use the multiple-choice answer sheet for Questions 1–10.

1. The function, $f(x)$, is one-to-one and has domain $[0,5)$ and range $(-2,5]$. 1

Which of the following correctly describes the domain and range of its inverse function, $f^{-1}(x)$?

	Domain	Range
(A)	$(-2,5]$	$[0,5)$
(B)	$[-2,5)$	$(0,5]$
(C)	$[0,5)$	$(-2,5]$
(D)	$(0,5]$	$[-2,5)$

2. The diagram below shows $y = x^2 + 6x - 6$ and $y = 2x - 1$. 1



Using the graph, or otherwise, for what values of x is $x^2 + 4x - 5 \leq 0$?

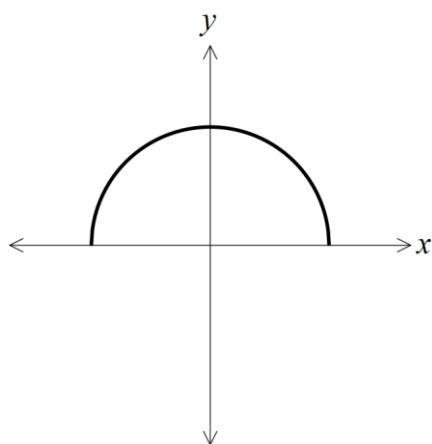
- (A) $[-11,1]$
(B) $(-\infty, -11] \cup [1, \infty)$
(C) $[-5,1]$
(D) $(-\infty, -5] \cup [1, \infty)$

3. What is the remainder when $x^4 - 3x^3 + x$ is divided by $(x + 1)$? 1
- (A) -5
(B) -1
(C) 1
(D) 3
4. Solve 1
- $$|x + 1| < 2$$
- (A) $-3 < x < 1$
(B) $x < -3$ or $x > 1$
(C) $-1 < x < 3$
(D) $x < -1$ or $x > 3$
5. There are 500 students in a school. 1
- Which of the following MUST be correct?
- (A) There is at least 1 month where at least 41 students have their birthday.
(B) There are at least 2 months where at least 41 students have their birthday.
(C) There is at least 1 month where at least 42 students have their birthday.
(D) There are at least 2 months where at least 42 students have their birthday.
6. What is the coefficient of the term containing x^2 in the expansion of $\left(x - \frac{1}{x}\right)^8$? 1
- (A) $\binom{8}{2}$
(B) $-\binom{8}{2}$
(C) $\binom{8}{5}$
(D) $-\binom{8}{5}$

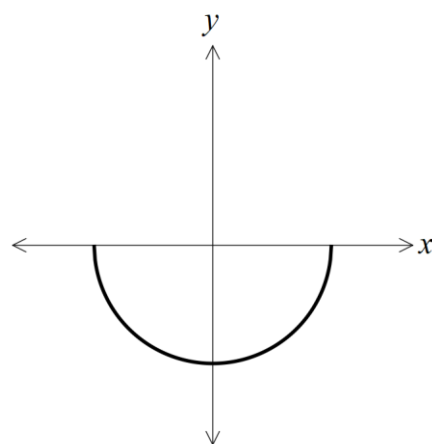
7. A curve on the xy -plane is defined parametrically by $x = \sin \theta$ and $y = \cos \theta$. 1

Which of the following best represents the curve for $0 \leq \theta \leq \pi$?

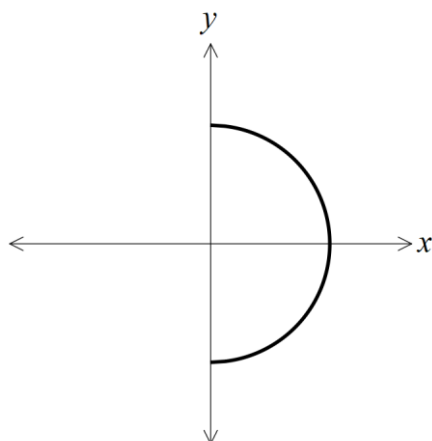
(A)



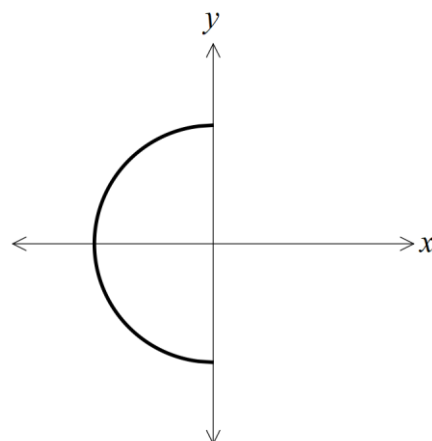
(B)



(C)



(D)



8. Let $P(x) = ax^3 + bx^2 + bx + a$ where a and b are constants and $a \neq 0$. 1

The three distinct zeroes are -1 , α and β .

Which of the following is a true statement?

- (A) $\beta = -\frac{1}{\alpha}$
(B) $\beta = \frac{1}{\alpha}$
(C) $\beta = \alpha - 1$
(D) $\beta = 1 - \alpha$

9. In the school playground, 2 girls and 4 boys are seated in a circle. How many arrangements are possible if the girls *cannot* sit next to each other? 1
- (A) 18
- (B) 24
- (C) 60
- (D) 72
-
10. The polynomial $P(x)$ has degree 5 and $P(x)$ is an odd function. 1
- Which of the following *could* be true?
- I: The sum of the real roots of $P(x)$ is positive.
II: $P(x) = 0$ has one distinct solution.
III: $P(x)$ has a double root.
- (A) I and II
- (B) II only
- (C) II and III
- (D) I and III

End of Section I

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Start a new writing booklet for each question. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

- (a) Suppose $P(x) = x^3 - 4x^2 + x + 6$.
- (i) Show that $(x + 1)$ is a factor of $P(x)$. **1**
 - (ii) Hence, write $P(x)$ as a product of linear factors. **2**
- (b) A particle's displacement in metres after t seconds is given by $x = t^2 - 4t + 3$.
- (i) Show that the particle is at rest after 2 seconds. **2**
 - (ii) Sketch the particle's displacement-time graph. **1**
 - (iii) What is the total distance travelled by the particle during the first 5 seconds? **1**
- (c) Solve $\frac{x}{x+1} \leq 2$. **3**
- (d) A curve in the xy -plane is defined parametrically by $x = 8t$ and $y = 4t^2$, where t is the parameter.
- (i) Draw the curve for $-3 \leq t \leq 3$, showing the co-ordinates of the endpoints and any intercepts. **2**
 - (ii) For what value(s) of t will the curve intersect the line $y = 10 - \frac{3}{4}x$? **3**

End of Question 11

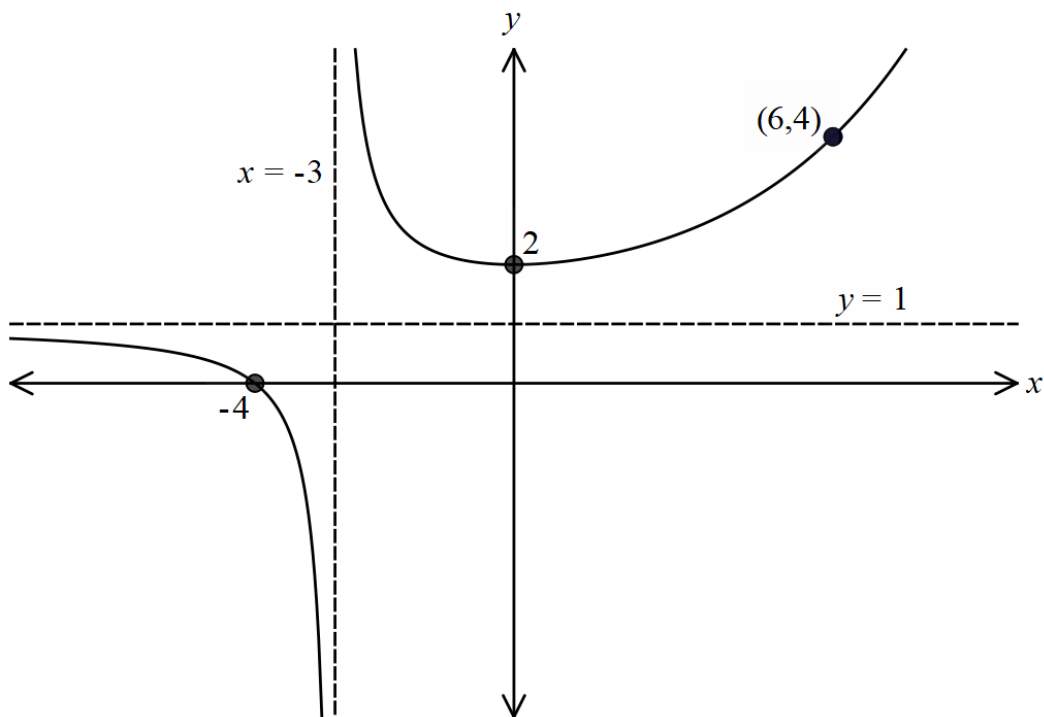
Question 12 (15 marks) Start a new writing booklet

- (a) (i) In how many ways can the letters in the word CAMPING be arranged? **1**
- (ii) In how many ways can the letters in the word EXAMINATION be arranged if the X must come first? **2**
- (b) Simplify **2**
- $$\frac{\cos 3x - \cos 5x}{\sin 3x + \sin 5x}$$
- (c) The function, $f(x) = e^{x^3}$, is one-to-one for all real values of x . **2**
- Find the equation of the inverse function, $f^{-1}(x)$.
- (d) A large ice cube is melting in such a way that its surface area, A , is decreasing at a rate of 6 cm^2 per minute.
- (i) Show that the side length of the cube, x , decreases at a rate of $\frac{1}{2x} \text{ cm/min}$. **2**
- (ii) At what rate is the side length decreasing when the ice cube's surface area is 600 cm^2 ? **2**
- (e) Neatly sketch $y = 3 \cos^{-1}\left(\frac{x}{2}\right)$ over its natural domain. **2**
- (f) The polynomial $P(x) = x^3 - kx^2 - (k + 1)x$ has a double root at $x = -1$. **2**
- Find the possible value(s) of k .

End of Question 12

Question 13 (15 marks) Start a new writing booklet

(a) Consider the graph of $y = f(x)$ below.



Answer parts (i) – (iii) on the **Q13a Insert** provided. Place the **Insert** inside your Question 13 writing booklet.

Sketch the following functions, showing all relevant information.

(i) $y = |f(x)|$ 1

(ii) $y = \sqrt{f(x)}$ 2

(iii) $y = \frac{1}{f(x)}$ 2

(b) In a football season with sixteen teams, there are 121 games. Each game is played between two teams. 2

Show that some pair of teams must have played each other more than once.

Question 13 continues on page 9

- (c) An island is found to be contaminated with the substance strontium-90 after an accident at its nuclear power plant. The rate at which the amount in kilograms, A , of strontium-90 decays t years after the accident is given by

$$\frac{dA}{dt} = -kA$$

where k is a constant.

- (i) Show that $A = A_0 e^{-kt}$ satisfies the differential equation, where A_0 is the amount of strontium-90 contaminating the island immediately after the accident. 1

- (ii) 29 years after the accident, scientists found that the amount of strontium-90 contaminating the island had halved. 2

Show that $k = \frac{1}{29} \ln 2$.

- (iii) The scientists declare that the island will become inhabitable again when the amount of strontium-90 falls below 5% of the initial amount. 2

How many years after the accident will the island become inhabitable?

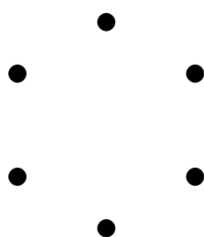
- (d) Given that $\sin \theta = \frac{1}{4}$ and $\cos \theta < 0$, find the exact value of $\tan 2\theta$. 3

End of Question 13

Question 14 (15 marks) Start a new writing booklet

- (a) (i) Consider the dots below.

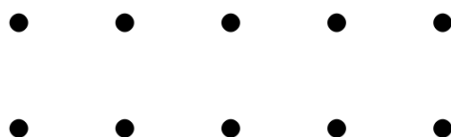
1



Explain why 6C_3 distinct triangles can be formed using the dots as vertices.

- (ii) How many distinct triangles can be formed using the dots below as vertices?

2



- (b) Show that $\sin(2 \arccos x) = 2x\sqrt{1-x^2}$ for $-1 \leq x \leq 1$.

2

- (c) Joe is studying a superconductor that has been cooled to its critical temperature. Unfortunately, he does not know the critical temperature because his laser thermometer cannot measure temperatures below -50°C .

4

Luckily, Joe knows that the superconductor will warm according to the differential equation

$$\frac{dT}{dt} = k(B - T)$$

which has a solution, $T = B - Ae^{-kt}$ (Do NOT prove this), where B is the temperature of the surrounding environment, A and k are constants and T is the temperature in $^\circ\text{C}$ after t minutes.

He places the superconductor, already at its critical temperature, in a 25°C laboratory. After half an hour, the laser thermometer shows that the temperature of the superconductor is -35°C . After a further half hour, its temperature is 5°C . What is the critical temperature of the superconductor?

Question 14 continues on page 11

- (d) (i) If $t = \tan \theta$, show that 3

$$\tan 4\theta = \frac{4t(1 - t^2)}{1 - 6t^2 + t^4}$$

- (ii) Given that $\theta = \frac{\pi}{10}$ and $\frac{3\pi}{10}$ are the roots of $\tan 4\theta = \cot \theta$ for $0 < \theta < \frac{\pi}{2}$, 3
find the exact value of $\tan \frac{\pi}{10}$.

End of paper

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Section I

① $f(x)$ has $D: [0, 5)$ and $R: (-2, 5]$

So $f^{-1}(x)$ has $D: (-2, 5]$ and $R: [0, 5)$ $\rightarrow$ (A)

② $x^2 + 4x - 5 \leq 0$

$$x^2 + 6x - 6 - 2x + 1 \leq 0$$

$$x^2 + 6x - 6 \leq 2x - 1$$

$y = x^2 + 6x - 6$ is below/on $y = 2x - 1$ for $-5 \leq x \leq 1 \rightarrow$ (C)

③ If $p(x) = x^4 - 3x^3 + x$ is divided by $(x+1)$,
by the Rem. Theorem, the remainder is

$$p(-1) = (-1)^4 - 3(-1)^3 + (-1)$$

$$= 1 + 3 - 1$$

$$= 3 \rightarrow \text{(D)}$$

④ $|x+1| < 2$

$$-2 < x+1 < 2$$

$$-3 < x < 1 \rightarrow \text{(A)}$$

⑤ 500 pigeons into 12 (months) holes:

$$\frac{500}{12} = 41.6... \quad (500 = 41 \times 12 + 8)$$

$\therefore$ At least one pigeonhole must contain
at least 42 pigeons $\rightarrow$ (C)

⑥ $[x + (-x^{-1})]^8$ has general term:

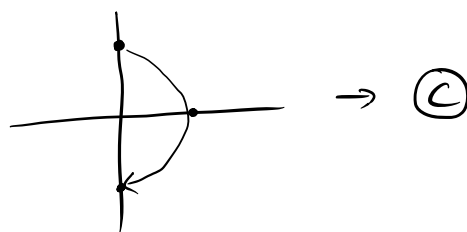
$${}^8C_r x^{8-r} (-x^{-1})^r = {}^8C_r x^{8-r} (-1)^r x^{-r} \\ = (-1)^r {}^8C_r x^{8-2r}$$

let $8-2r=2 \rightarrow r=3:$

$$= (-1)^3 {}^8C_3 x^2 \\ = \boxed{-{}^8C_3} x^2 \rightarrow \textcircled{D}$$

⑦

θ	0	$\frac{\pi}{2}$	π
x	0	1	0
y	1	0	-1
	$\downarrow$	$\downarrow$	$\downarrow$
	(0,1)	(1,0)	(0,-1)



⑧

$$p(x) = ax^3 + bx^2 + cx + a$$

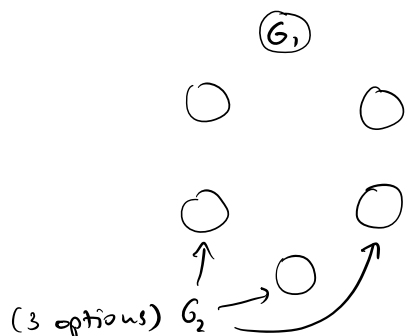
let zeroes be $\alpha, \beta, -1$

Product of zeroes: $\alpha \cdot \beta \cdot (-1) = -\frac{a}{a}$

$$-\alpha\beta = -1 \\ \beta = \frac{1}{\alpha} \rightarrow \textcircled{B}$$

⑨ $\boxed{1} \times \boxed{3} \times \boxed{4!} = 72 \rightarrow \textcircled{D}$

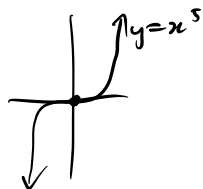
Place 1st girl Place 2nd girl Arrange boys



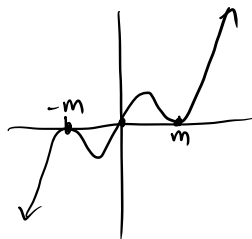
⑩ X I: For each positive root of $P(x)$, there must exist an opposite negative root to preserve point symmetry

$\therefore$ The sum of roots must equal zero.

✓ II: E.g. $x^5 = 0$ has one distinct soln, $x = 0$



✓ III: E.g.



$$P(x) = x(x-m)^2(x+m)^2$$

So $\textcircled{C}$

Section II

Question 11

(a) $P(x) = x^3 - 4x^2 + x + 6$

(i) $P(-1) = (-1)^3 - 4(-1)^2 + (-1) + 6$
 $= -1 - 4 - 1 + 6$
 $= 0$

$\therefore$ By Factor Theorem $(x+1)$ is a factor of $P(x)$.

(ii)
$$\begin{array}{r} x^2 - 5x + 6 \\ x+1 \overline{) x^3 - 4x^2 + x + 6} \\ \underline{x^3 + x^2} \\ -5x^2 + x \\ \underline{-5x^2 - 5x} \\ 6x + 6 \\ \underline{6x + 6} \\ 0 \end{array} \quad \begin{aligned} &\rightarrow P(x) = (x+1)(x^2 - 5x + 6) \\ &\therefore P(x) = (x+1)(x-2)(x-3) \end{aligned}$$

(b) (i) 'at rest' $\rightarrow v = 0$

$$x = t^2 - 4t + 3$$

$$v = \frac{dx}{dt} = 2t - 4$$

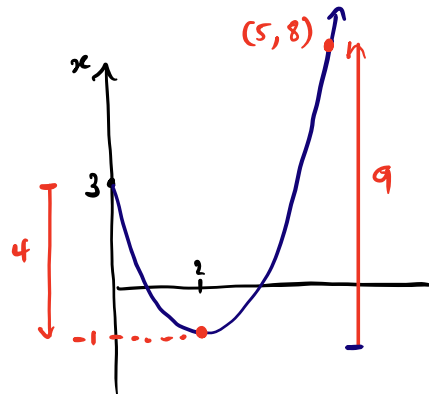
$$\text{let } 2t - 4 = 0$$

$$2t = 4$$

$$t = 2$$

$\therefore$ The particle is at rest after 2 seconds.

(ii)



(iii) Consider endpoints and turning pts:

$$\begin{aligned} \text{at } t=0, x &= 3 \\ t=2, x &= 2^2 - 4(2) + 3 \\ &= -1 \\ t=5, x &= 5^2 - 4(5) + 3 \\ &= 8 \end{aligned}$$

$$\begin{aligned} \text{Total distance} &= 4 + 9 \\ \text{travelled} &= 13 \text{ m.} \end{aligned}$$

(c) $\frac{x}{x+1} \leq 2$ (Note $x \neq -1$)

$$x(x+1) \leq 2(x+1)^2 \quad (\text{Multiplying by } (x+1)^2)$$

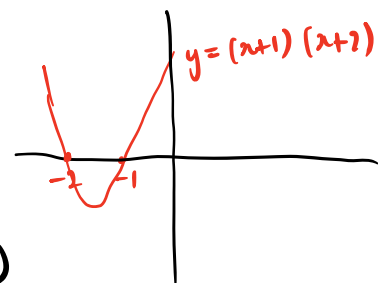
$$x(x+1) - 2(x+1)^2 \leq 0$$

$$(x+1)[x - 2(x+1)] \leq 0$$

$$(x+1)(x - 2x - 2) \leq 0$$

$$(x+1)(-x-2) \leq 0$$

$$(x+1)(x+2) \geq 0 \quad (\text{Dividing by } -1)$$

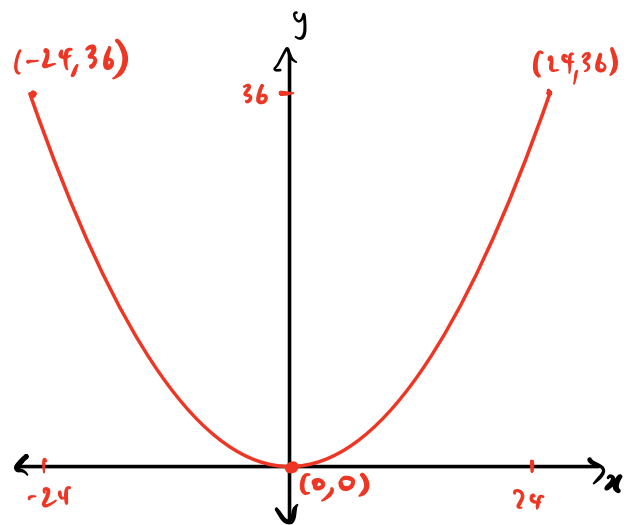


$$\therefore x \leq -2, x > -1 \quad (x \neq -1)$$

(d) $x = 8t, y = 4t^2$

(i) $\frac{x}{4} = 2t, y = (2t)^2$
 $y = \left(\frac{x}{4}\right)^2$
 $\therefore y = \frac{1}{16} x^2$

t	-3	0	3
x	-24	0	24
y	36	0	36



(ii) $y = 10 - \frac{3}{4}x$

Sub $x = 8t, y = 4t^2$:

$$4t^2 = 10 - \frac{3}{4}(8t)$$

$$4t^2 = 10 - 6t$$

$$4t^2 + 6t - 10 = 0$$

$$2t^2 + 3t - 5 = 0$$

$$2t^2 - 2t + 5t - 5 = 0$$

$$2t(t-1) + 5(t-1) = 0$$

$$(t-1)(2t+5) = 0$$

$$t = 1, -\frac{5}{2}$$

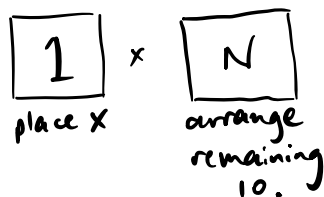
$\therefore y = 6 - 2x$ and the curve intersect for $t = 1$ and $t = -\frac{5}{2}$.

Question 12

(a) (i) 7 distinct objects

= 7! ways or 5040 ways

(ii)



→ 10 objects, 3 repeated pairs:
A's, I's, N's

$$\rightarrow N = \frac{10!}{2!2!2!} = 453600$$

$\therefore$ There are 453600 ways.

$$\begin{aligned} \text{(b)} \quad \frac{\cos 3x - \cos 5x}{\sin 3x + \sin 5x} &= \frac{\cos(\overset{A}{4x} - \overset{B}{x}) - \cos(\overset{A}{4x} + \overset{B}{x})}{\sin(\overset{A}{4x} - \overset{B}{x}) + \sin(\overset{A}{4x} + \overset{B}{x})} \\ &= \frac{2 \sin 4x \sin x}{2 \sin 4x \cos x} \quad (\text{from Ref. Sheet}) \\ &= \tan x \end{aligned}$$

$$\text{(c)} \quad f: y = e^{x^3}$$

$$f^{-1}: x = e^{y^3}$$

$$y^3 = \ln x$$

$$y = \sqrt[3]{\ln x}$$

$$\therefore f^{-1}(x) = \sqrt[3]{\ln x}$$

(d) (i) $\frac{dA}{dt} = -6$

$A = 6x^2$ (6 square feet)

$\frac{dA}{dx} = 12x$

$$\frac{dx}{dt} = \frac{dx}{dA} \times \frac{dA}{dt}$$

$$= \frac{1}{12x} \times -6$$

$\therefore \frac{dx}{dt} = -\frac{1}{2x} \rightarrow \therefore$ The side length is decreasing at a rate of $\frac{1}{2x}$ cm/min.

(ii) If $A = 600$

$6x^2 = 600$

$x^2 = 100$

then $x = 10$ cm (> 0)

If $x = 10$, $\frac{dx}{dt} = -\frac{1}{2(10)}$

$= -\frac{1}{20} \rightarrow \therefore$ When $A = 600 \text{ cm}^2$, the side length is decreasing at $\frac{1}{20}$ cm/min.

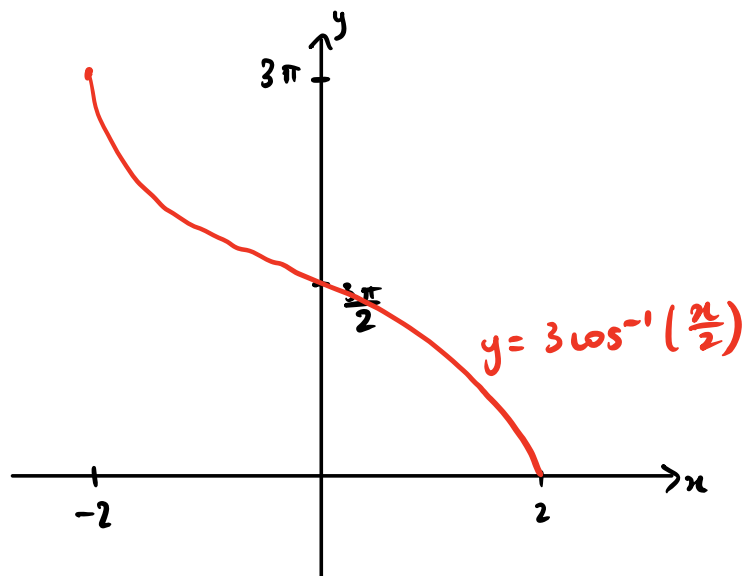
(e) $y = 3 \cos^{-1}\left(\frac{x}{2}\right)$

D: $-1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$

R: $0 \leq \cos^{-1}\left(\frac{x}{2}\right) \leq \pi$

$0 \leq 3 \cos^{-1}\left(\frac{x}{2}\right) \leq 3\pi$



(f) If $x = -1$ is a double root of $P(x)$, then:

$$P(-1) = P'(-1) = 0$$

$$P(-1) = (-1)^3 - k(-1)^2 - (k+1)(-1)$$

$$= -1 - k + k + 1$$

$$= 0 \quad (\text{not helpful to find } k)$$

$$P'(x) = 3x^2 - 2kx - (k+1)$$

$$P'(-1) = 3(-1)^2 - 2k(-1) - k - 1$$

$$0 = 3 + 2k - k - 1$$

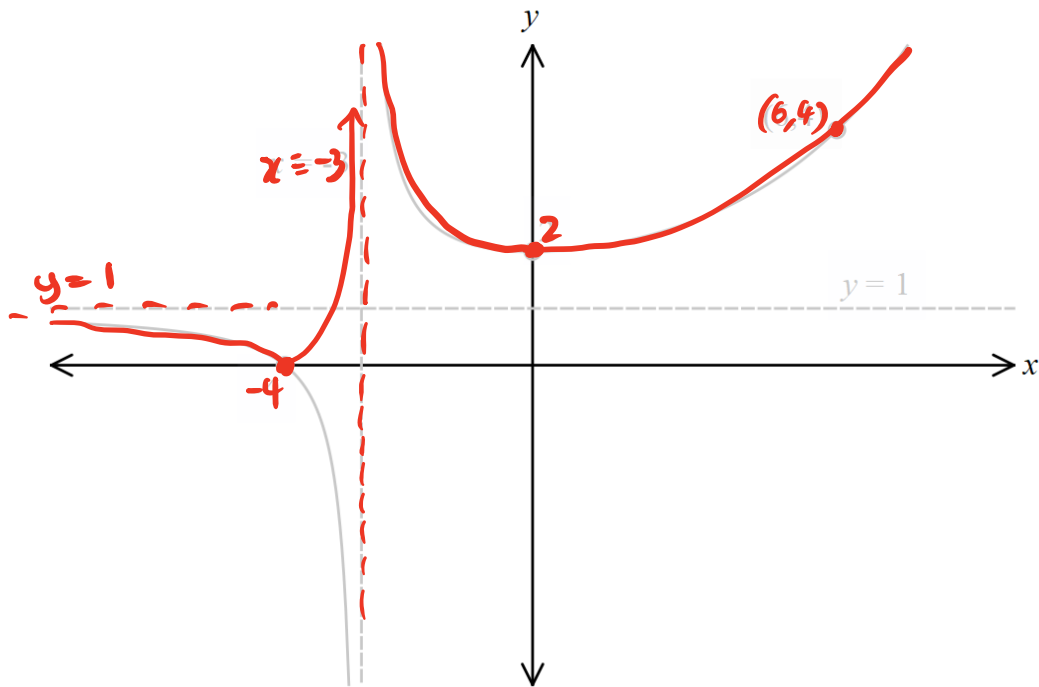
$$0 = 2 + k$$

$$\therefore k = -2$$

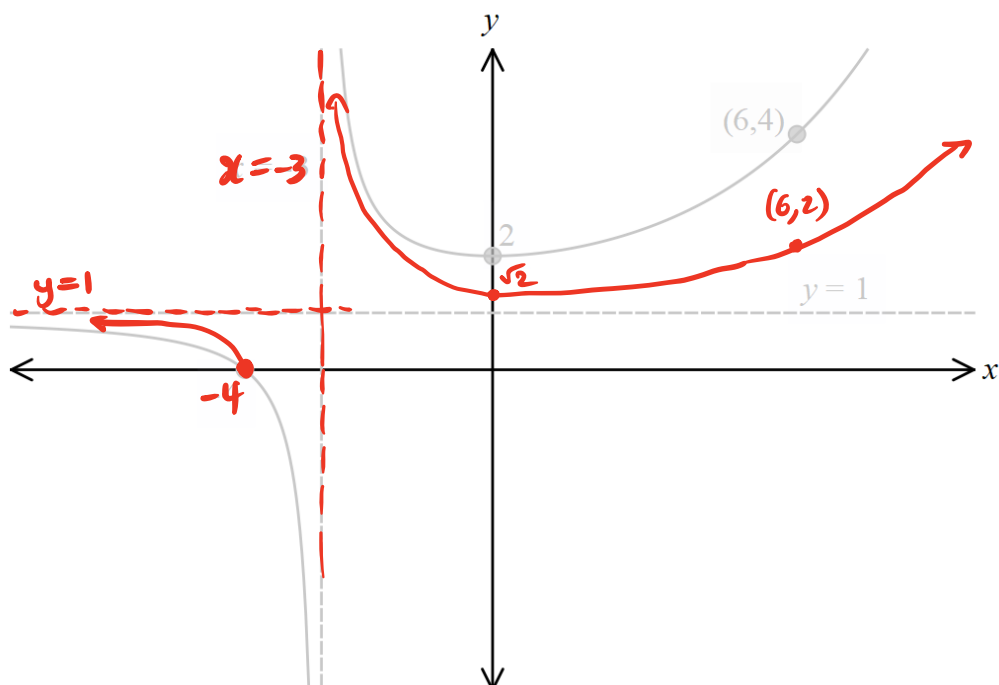
Q13a Insert

Note: If the transformed function shares features with $y = f(x)$, these features should be redrawn/relabelled.

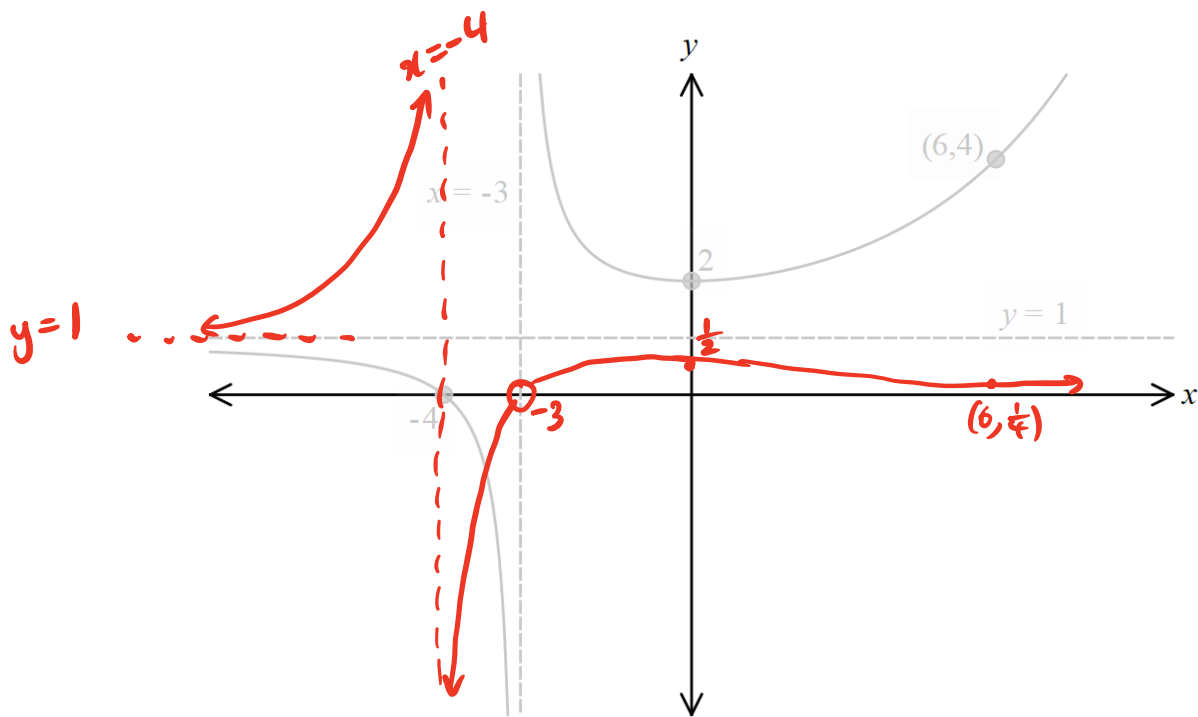
(i) $y = |f(x)|$



(ii) $y = \sqrt{f(x)}$



(iii) $y = \frac{1}{f(x)}$



Write your Assessment Number on the front page of this insert and place in your Question 13 writing booklet.

Question 13

- (b) Ways of selecting 2 teams from 16 = ${}^{16}C_2 = 120$
(no repeat fixtures)

Placing 121 pigeons into 120 holes
(# games) (distinct games)

means one hole must contain at least two pigeons.

$\therefore$ At least one pair of teams will play each other more than once.

(c) (ii) $A = A_0 e^{-kt}$

$$\frac{dA}{dt} = -k A_0 e^{-kt}$$

$$\frac{dA}{dt} = -k A$$

$\therefore A = A_0 e^{-kt}$ satisfies the differential equation.

(ii) at $t = 29$, $A = \frac{1}{2} A_0$:

$$\frac{1}{2} A_0 = A_0 e^{-29k}$$

$$-29k = \ln \frac{1}{2}$$

$$-29k = -\ln 2$$

$$\therefore k = \frac{1}{29} \ln 2$$

(iii) let $A < 0.05 A_0$

$$A_0 e^{-kt} < 0.05 A_0 \quad (A_0 > 0)$$

$$\ln(e^{-kt}) < \ln 0.05$$

$$-kt < \ln 0.05$$

$$t > \frac{\ln 0.05}{-\frac{1}{29} \ln 2} \quad (k > 0 \text{ so } -k < 0)$$

$$t > 125.3 \dots$$

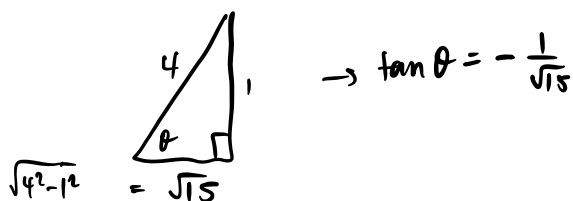
$\therefore$ The island will be inhabitable after 126 years (next whole year).

(d) $\sin \theta = \frac{1}{4} > 0$, $\cos \theta < 0$

S	A
T	C

 $\therefore \tan \theta < 0$

With $\sin \theta = \frac{1}{4}$



$$\begin{aligned} \tan 2\theta &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\ &= \frac{2 \left(-\frac{1}{\sqrt{15}}\right)}{1 - \left(-\frac{1}{\sqrt{15}}\right)^2} \\ &= \frac{-\frac{2}{\sqrt{15}}}{1 - \frac{1}{15}} \\ &= \frac{-\frac{2}{\sqrt{15}}}{\frac{14}{15}} \\ &= -\frac{15}{7\sqrt{15}} \times \frac{\sqrt{15}}{\sqrt{15}} \\ &= -\frac{\sqrt{15}}{7} \end{aligned}$$

Question 14

- (i) Each triangle is formed by choosing 3 distinct dots as vertices from 6 options.
6 options, choose 3 $\rightarrow {}^6C_3$ triangles.
- (ii) Degenerate triangles must be ignored, such as the 'triangle' formed by the choice of these points:



Method 1

Each triangle must be formed from two dots in one row and one in the other:

$$\boxed{{}^2C_1} \times \boxed{{}^5C_2} \times \boxed{{}^5C_1} = 100 \text{ triangles.}$$

choose row for 2 dots choose 2 dots choose a dot in the other row

Method 2

Remove degenerate triangles:

$$\boxed{{}^2C_1} \times \boxed{{}^5C_3} = 20$$

choose row for deg. Δ choose 3 dots

$${}^{10}C_3 - 20 = 100 \text{ triangles.}$$

total choices of 3 dots

(b) let $y = \cos^{-1}x$

$$\sin(2 \cos^{-1}x) = \sin 2y$$

$$= 2 \sin y \cos y$$

If $y = \cos^{-1}x$, $x = \cos y$



so $\sin y = \sqrt{1-x^2}$

so $2 \sin y \cos y = 2 \cdot \sqrt{1-x^2} \cdot x$

$$\therefore \sin(2 \cos^{-1}x) = 2x \sqrt{1-x^2}.$$

(c) $B = 25$ (surrounding environment)

at $t=30$, $T=-35 \rightarrow -35 = 25 - Ae^{-30k}$

$$Ae^{-30k} = 60 \quad (1)$$

at $t=60$, $T=5 \rightarrow 5 = 25 - Ae^{-60k}$

$$Ae^{-60k} = 20 \quad (2)$$

$$(1)^2 \div (2) : \frac{A^2 e^{-60k}}{A e^{-60k}} = \frac{3600}{20}$$

$$A = 180$$

at $t=0$,

$$T = B - Ae^0$$

$$= B - A$$

$$= 25 - 180$$

$$= -155^\circ\text{C}$$

$\therefore$ The critical temperature of the superconductor is -155°C .

(d) (i) If $t = \tan \theta$, $\tan 2\theta = \frac{2t}{1-t^2}$ (+ formula)

$$\text{LHS} = \tan 4\theta$$

$$= \tan (2 \cdot 2\theta)$$

$$= \frac{2 \tan 2\theta}{1 - \tan^2 2\theta}$$

$$= \frac{2 \cdot \frac{2t}{1-t^2}}{1 - \left(\frac{2t}{1-t^2}\right)^2}$$

$$= \frac{\frac{4t}{\cancel{1-t^2}}}{\frac{(1-t^2)^2}{\cancel{(1-t^2)^2}} - \frac{4t^2}{\cancel{(1-t^2)^2}}} \times \frac{(1-t^2)^2}{(1-t^2)^2}$$

$$= \frac{4t(1-t^2)}{(1-t^2)^2 - 4t^2}$$

$$= \frac{4t(1-t^2)}{1-2t^2+t^2-4t^2}$$

$$= \frac{4t(1-t^2)}{1-6t^2+t^4}$$

$$= \text{RHS}$$

(ii) $\theta = \frac{\pi}{10}$ and $\frac{3\pi}{10}$ are solutions to $\tan 4\theta = \cot \theta$

If $t = \tan \theta$, the equation becomes:

$$\frac{4t(1-t^2)}{1-6t^2+t^4} = \frac{1}{t} \quad \left(\text{from (i) and } \cot \theta = \frac{1}{\tan \theta} \right)$$

$$4t^2(1-t^2) = 1-6t^2+t^4$$

$$4t^2 - 4t^4 = 1-6t^2+t^4$$

$$5t^4 - 10t^2 + 1 = 0$$

Using quadratic formula:

$$t^2 = \frac{10 \pm \sqrt{100 - 4(5)(1)}}{10}$$

$$t^2 = \frac{10 \pm \sqrt{80}}{10}$$

$$t^2 = \frac{10 \pm 4\sqrt{5}}{10} = \frac{5 \pm 2\sqrt{5}}{5}$$

$$t = \pm \sqrt{\frac{5 \pm 2\sqrt{5}}{5}}$$

Since $\tan \frac{\pi}{10}$, $\tan \frac{3\pi}{10} > 0$ (1st Quad), ignore negative values:

$$t = \sqrt{\frac{5 \pm 2\sqrt{5}}{5}}$$

$$t = \sqrt{\frac{5-2\sqrt{5}}{5}} \quad \text{or} \quad \sqrt{\frac{5+2\sqrt{5}}{5}}$$

Since $\tan \theta$ is increasing, the smaller value corresponds to the solution $\tan \frac{\pi}{10}$.

$$\therefore \tan \frac{\pi}{10} = \sqrt{\frac{5-2\sqrt{5}}{5}}$$